

Agenda

File Handling
Exception Handling
Modules Builtin & third party
AI assisted Coding

Word & line counter

1. Take a filename as input
2. read the file
3. Count the number of characters, lines, words, vowels & punctuations
4. Create a summary.log file
5. Write the summary to it. Next time it should append to the summary log

Sample Output:

filename: data.csv

Number of Lines: 4
Number of Words: 14
Number of characters: 81

Built-in modules

os - Files & os functions

sys - system tasks

pathlib - helps manipulate paths

urllib - handling internet traffic

zip/shutil - zipping the files and folders

venv - virtual environment

third party modules

requests - internet interaction

numpy - numerical computing

pandas - python data science

matplotlib - charts & visualization

altair charts - dynamic charts

reportlab / fpdf - pdf report generation

ppt2md - ppt to markdown

BeautifulSoup - HTML Parsing

sklearn - ML Models

keras - AI Models

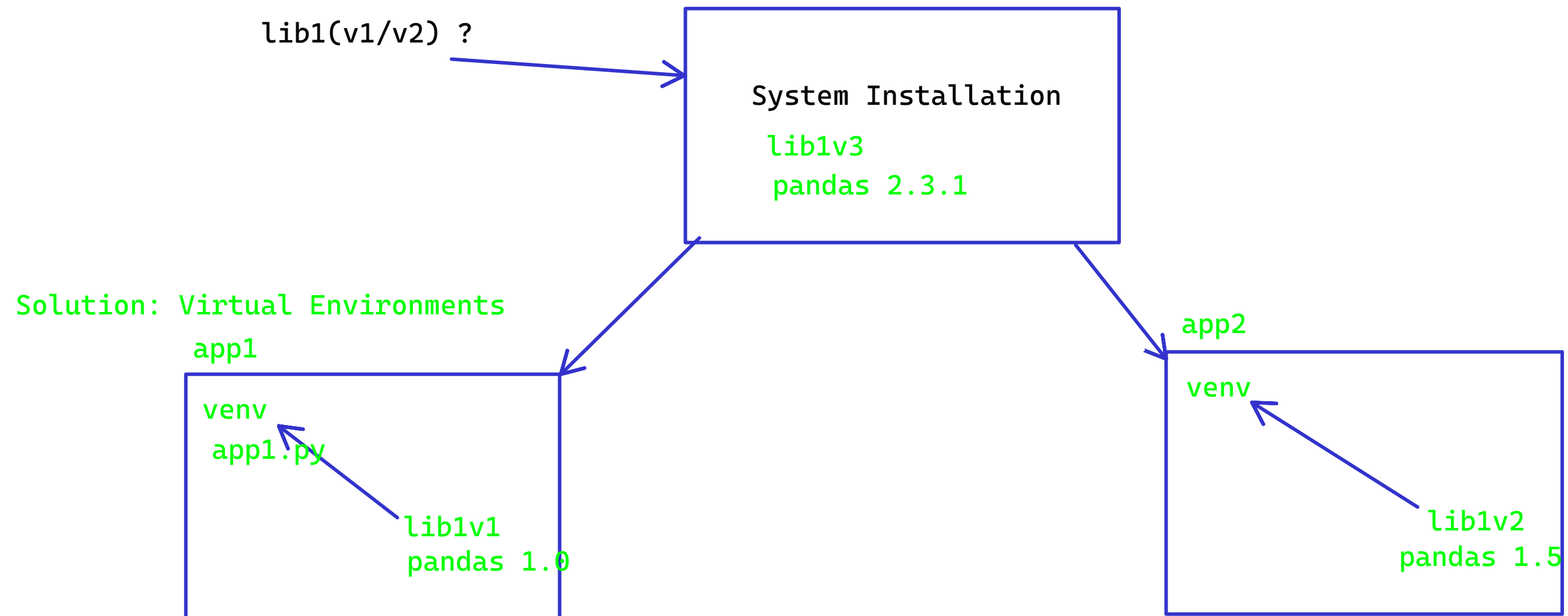
langchain/langgraph/opencrew - AI Agents

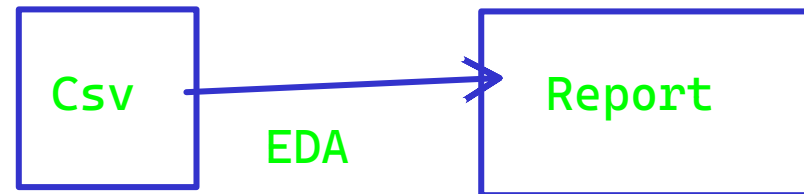
PIL - Python Imaging library

OpenCV - Computer Vision

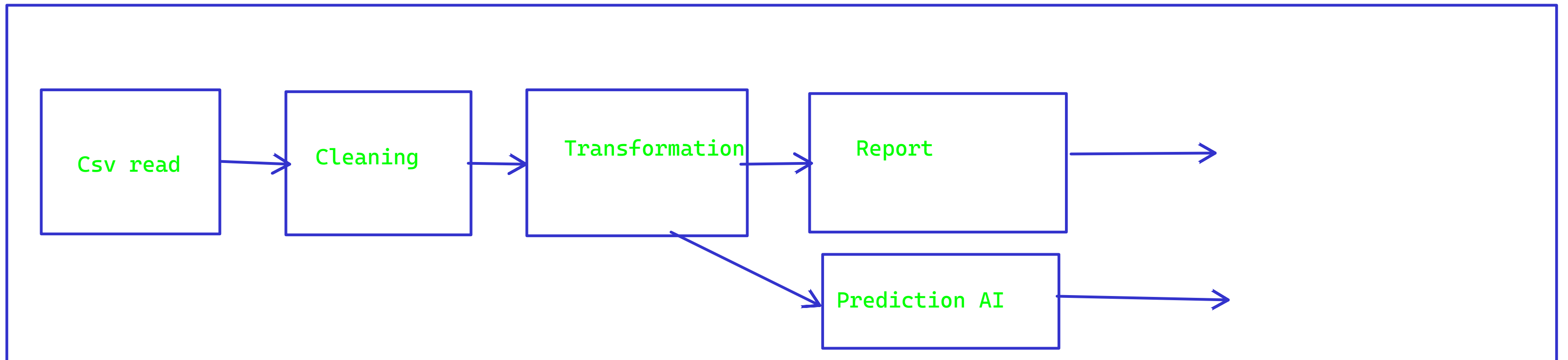
venv - Virtual Environments

Problem: Assume you are working on multiple projects, app1 depends on lib1v1 & app2 depends on lib1v2





Pipeline / ETL



Capstone Process

